

INTERMEDIATE ALGEBRA — INSTRUCTOR COPY

Chapters 1 & 2 Quiz: Solutions

Full worked solutions, with the source problem cited for each question

Grading note. Each question is worth 10 points. The boxed value is the answer; the lines above it are the work students are expected to show. Where a problem has a standard trap — an extraneous root, a strict versus non-strict domain endpoint, an excluded value in a range — it is flagged in *italics* so partial credit can be applied consistently.

Part I — Chapter 1

Basic Techniques for Solving Equations

1. *Review Problem 1.23(b), §1.1*Solve $x - 1 \leq 3x + 2 \leq 2x + 6$; write the answer in interval notation.

| A double inequality is two inequalities at once; solve each separately and intersect.

Left: $x - 1 \leq 3x + 2$. Subtracting x from both sides gives $-1 \leq 2x + 2$, so $-3 \leq 2x$, and $x \geq -\frac{3}{2}$.Right: $3x + 2 \leq 2x + 6$. Subtracting $2x$ gives $x + 2 \leq 6$, so $x \leq 4$.

Both must hold, so the solution is the overlap.

$$\boxed{\left[-\frac{3}{2}, 4\right]}$$

Note: no multiplication or division by a negative occurs here, so no inequality sign flips. Both endpoints are included because both inequalities are non-strict.

2. *Review Problem 1.19(a), §1.3*Solve the system $4x + 5y = 43$, $9x - 2y = 57$.| Eliminate y : multiply the first equation by 2 and the second by 5 so the y -coefficients become +10 and -10.

$$8x + 10y = 86,$$

$$45x - 10y = 285.$$

Adding gives $53x = 371$, so $x = 7$. Substituting into $4x + 5y = 43$ gives $28 + 5y = 43$, so $5y = 15$ and $y = 3$.Check in the second original equation: $9(7) - 2(3) = 63 - 6 = 57$. ✓

$$\boxed{(x, y) = (7, 3)}$$

3. *Review Problem 1.20, §1.2*

Jeff is 4 times older than his daughter. Five years ago he was 9 times older than his daughter. How old is his daughter?

| Let d be the daughter's age now; then Jeff's age is $4d$. Five years ago their ages were $d - 5$ and $4d - 5$, and the second was 9 times the first:

$$4d - 5 = 9(d - 5).$$

Expanding gives $4d - 5 = 9d - 45$, so $40 = 5d$ and $d = 8$.Check: the daughter is 8 and Jeff is 32; five years ago they were 3 and 27, and $27 = 9 \cdot 3$. ✓
$$\boxed{\text{The daughter is 8 years old.}}$$

Note: students who set up $4d - 5 = 9d - 5$ (forgetting to age the daughter down as well) get $d = 0$; the check is what catches this.

4. Review Problem 1.24, §1.2

Find all ordered pairs (x, y) with $\sqrt{x} + \sqrt{y} = 7$ and $3\sqrt{x} - 4\sqrt{y} = -14$.

| The system is linear in $\sqrt{x}$ and $\sqrt{y}$, so substitute $u = \sqrt{x}$ and $v = \sqrt{y}$, where $u \geq 0$ and $v \geq 0$:

$$\begin{aligned}u + v &= 7, \\3u - 4v &= -14.\end{aligned}$$

From the first, $u = 7 - v$. Substituting into the second gives $3(7 - v) - 4v = -14$, so $21 - 7v = -14$, hence $v = 5$ and $u = 2$.

Both are non-negative, so both are legitimate square roots. Squaring, $x = u^2 = 4$ and $y = v^2 = 25$.

Check: $\sqrt{4} + \sqrt{25} = 2 + 5 = 7$ and $3(2) - 4(5) = -14$. ✓

$$(x, y) = (4, 25)$$

Note: a negative value of u or v would have to be discarded, since $\sqrt{}$ denotes the non-negative root. Here neither is negative, so exactly one ordered pair survives.

5. Review Problem 1.21(a), §1.4

Solve $8x + y - z = 46$, $3x + 4y - 2z = 27$, $4x - y + z = 14$.

| The first and third equations have opposite y - and z -coefficients, so adding them eliminates both variables at once:

$$(8x + y - z) + (4x - y + z) = 46 + 14 \implies 12x = 60,$$

so $x = 5$.

Substituting $x = 5$ into the third equation gives $20 - y + z = 14$, so $y - z = 6$.

Substituting $x = 5$ into the second gives $15 + 4y - 2z = 27$, so $4y - 2z = 12$, that is, $2y - z = 6$.

Subtracting $y - z = 6$ from $2y - z = 6$ gives $y = 0$, and then $z = y - 6 = -6$.

Check: $8(5) + 0 - (-6) = 46$ ✓, $3(5) + 0 - 2(-6) = 27$ ✓, $4(5) - 0 + (-6) = 14$ ✓.

$$(x, y, z) = (5, 0, -6)$$

Note: spotting that equations 1 and 3 add to eliminate two variables is the intended shortcut, but any correct elimination or substitution order earns full credit.

Part II — Chapter 2

Functions Review

6. Review Problem 2.23(d), §2.1

Find the domain and range of $S(x) = \frac{12x - 9}{6 - 9x}$.

| **Domain.** The only restriction is that the denominator cannot vanish: $6 - 9x = 0$ when $x = \frac{2}{3}$.

$$\text{Domain: } (-\infty, \frac{2}{3}) \cup (\frac{2}{3}, \infty)$$

Range. Use the chapter's strategy: set the function equal to a new variable and solve for x . Writing $y = \frac{12x - 9}{6 - 9x}$ and clearing the denominator,

$$y(6 - 9x) = 12x - 9 \implies 6y - 9xy = 12x - 9.$$

Collect the x terms on one side: $6y + 9 = 12x + 9xy = x(12 + 9y)$, so

$$x = \frac{6y + 9}{9y + 12}.$$

This produces a valid input x for every y except the one making the denominator zero: $9y + 12 = 0$, i.e. $y = -\frac{4}{3}$. That value of y is therefore never attained, and every other value is.

$$\text{Range: } \left(-\infty, -\frac{4}{3}\right) \cup \left(-\frac{4}{3}, \infty\right)$$

Note: $-\frac{4}{3}$ is the ratio of the leading coefficients, $12/(-9)$ — the horizontal asymptote. Answers of “all real numbers” for the range are the common error and should lose the range half of the credit.

7. Review Problem 2.24(a), §2.1

Find the domain of $f(x) = \frac{1}{\sqrt{2x-5}} + \sqrt{9-3x}$.

| Each term imposes its own restriction, and x must satisfy both.

First term: $\sqrt{2x-5}$ must be defined *and* nonzero, since it sits in a denominator. So $2x-5 > 0$, giving $x > \frac{5}{2}$ — a *strict* inequality.

Second term: $9-3x \geq 0$, giving $x \leq 3$.

Intersecting the two conditions:

$$\left(\frac{5}{2}, 3\right]$$

Note: the endpoint asymmetry is the whole point of the problem. Writing $\left[\frac{5}{2}, 3\right]$ (treating the first radical as if it were not in a denominator) is the expected error.

8. Review Problem 2.35, §2.2

How can the graph of $f(x)$ be used to produce the graph of $y = |f(x)|$?

| By the definition of absolute value,

$$|f(x)| = \begin{cases} f(x), & \text{if } f(x) \geq 0, \\ -f(x), & \text{if } f(x) < 0. \end{cases}$$

So wherever the graph of f is on or above the x -axis, the graph of $|f|$ is identical to it; wherever the graph of f is below the x -axis, the output changes sign, which reflects that piece across the x -axis.

Keep every part of the graph of f that lies on or above the x -axis, and reflect every part that lies below the x -axis across the x -axis.

Full credit requires both halves of the description plus a reason. The resulting graph never dips below the x -axis, and it meets the original graph exactly at the x -intercepts of f , where the two pieces join.

9. Review Problem 2.25, §2.3

With $f(x) = \frac{4x}{x+2}$ and $g(x) = \frac{2x}{x+4}$, find $f(g(x))$.

| Substitute $g(x)$ into f wherever x appears:

$$f(g(x)) = \frac{4 \cdot \frac{2x}{x+4}}{\frac{2x}{x+4} + 2} = \frac{\frac{8x}{x+4}}{\frac{2x}{x+4} + 2}.$$

Multiply the numerator and the denominator by $x + 4$ to clear the inner fractions:

$$f(g(x)) = \frac{8x}{2x + 2(x + 4)} = \frac{8x}{4x + 8}.$$

Dividing numerator and denominator by 4,

$$f(g(x)) = \frac{2x}{x + 2}$$

Note: strictly, this composition is undefined at $x = -4$ (outside the domain of g) and at $x = -2$ (where $g(x) + 2 = 0$), even though the simplified formula hides the first of these. Mentioning either restriction is a bonus, not a requirement.

10. Review Problem 2.37, §2.4 (Source: AHSME)

$f(x) = \frac{cx}{2x + 3}$ satisfies $f(f(x)) = x$ for all real x except $-\frac{3}{2}$. Find c .

| Compute the composition, multiplying numerator and denominator by $2x + 3$:

$$f(f(x)) = \frac{c \cdot \frac{cx}{2x + 3}}{2 \cdot \frac{cx}{2x + 3} + 3} = \frac{c^2x}{2cx + 3(2x + 3)} = \frac{c^2x}{(2c + 6)x + 9}.$$

Setting this equal to x and clearing the denominator gives

$$c^2x = x[(2c + 6)x + 9] = (2c + 6)x^2 + 9x,$$

so $(2c + 6)x^2 + (9 - c^2)x = 0$ must hold for every allowed x . A polynomial that is identically zero has all coefficients zero:

$$2c + 6 = 0 \quad \text{and} \quad 9 - c^2 = 0.$$

The first gives $c = -3$; the second gives $c = \pm 3$. Only $c = -3$ satisfies both.

$$c = -3$$

Check: with $f(x) = \frac{-3x}{2x + 3}$,

$$f(f(x)) = \frac{-3 \cdot \frac{-3x}{2x + 3}}{2 \cdot \frac{-3x}{2x + 3} + 3} = \frac{9x}{-6x + 6x + 9} = \frac{9x}{9} = x. \checkmark$$

Note: f being its own inverse is what $f(f(x)) = x$ says. Students who substitute a single convenient value of x (say $x = 1$) can find a candidate for c but have not shown the identity holds for all x ; that is worth partial credit only if they verify the candidate afterwards, as above.

All problems are taken from Richard Rusczyk, David Patrick, and Ravi Boppana, Art of Problem Solving: Intermediate Algebra, Chapters 1–2 (Review Problems). Every answer above was verified symbolically with SymPy.